BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

CHEMISTRY (PART-I) — (Fresh/Re-appear) — (Version-B) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 18 | **Paper Code:** BN-XI-XXIII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Hydrogen gas molecules would have no kinetic energy at

(A) -273°C
(B) 0°C
(C) 25°C
(D) 273°C

Q2. Which energy level allows only absorption and not emission of energy

(A) 3d
(B) 2p
(C) 2s
(D) 1s

Q3. Bohr's atomic model fails to explain the spectrum of

(A) H^{+1}
(B) He^{+1}
(C) Li^{+2}
(D) Be^{+3}

Q4. In pressure cooker, the boiling point of liquid is

(A) Constant
(B) Zero
(C) Increased
(D) Decreased

Q5. A limiting reactant in the following reaction is $H_2 + Cl_2 \longrightarrow 2HCl$

(A) H_2
(B) Cl_2
(C) HCl
(D) None

Q6. Diamond and graphite are

(A) Isomorphs
(B) Isotopes
(C) Allotropes
(D) Isotones

Q7. At 25°C, which gas will show maximum deviation from ideal behaviour

(A) CH_4
(B) N_2
(C) O_2
(D) NH_3

Q8. Magnetic quantum number explains

(A) Stark effect
(B) Zeeman effect
(C) Tunnel effect
(D) None of these

Q9. The least acidic of the following is

(A) HI
(B) HBr
(C) HCl
(D) HF

Q10. What is the mass (in grams) of 5 moles of water (H_2O)

(A) 90g
(B) 36g
(C) 18g
(D) 100g

Q11. pH of an aquous solution is 9. It pOH is:

(A) 11
(B) 9
(C) 7
(D) 5

Q12. No work is done at constant:

(A) Pressure
(B) Volume
(C) Temperature
(D) Mass

Q13. Salt hydrolysis is the reverse of

(A) Neutralization
(B) Hydration
(C) Desalination
(D) Esterification

Q14. Comparatively more dilute aqueous solution of the following is

(A) 1M NaOH
(B) 1N NaOH
(C) 1m NaOH
(D) 1.5M NaOH

Q15. When moles of reactants and products are equal, then

(A) Kp =Kc
(B) Kp> Kc
(C) Kp< Kc
(D) Kp≥ Kc

Q16. A buffer solution that completely neutralizes the added strong acid, is

(A) Acidic (B) Basic

(C) Neutral (D) None

Q17. A binary ionic compound of greater lattice energy is

(A) NaCl (B) KBr

(C) KI (D) LiF

Q18. The angle between two un-hybrid p-orbitals of the same atom / ion is

(A) Zero (B) 90°

(C) 109.5° (D) 120°

CHEMISTRY (PART-I) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40Hrs | **Maximum Marks:** 40

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 40) Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. Define resonance. What are the conditions for writing resonating structure?

ii. What is electro chemical series and its significance?

iii. Calculate the molality of solution which contains 80 grams NaOH dissolved in 1000 grams of water.

iv. Define evaporation. Why evaporation causes cooling?

v. What are London forces? Give their significance

vi. Calculate the frequency of a radiation which has wave number $0.5\times10^{8}m^{-1}$

vii. What is gram-atoms? Why the concept of gram-atoms is useful in chemistry?

viii. What is meant by Joule Thompson Effect?

ix. Calculate pH of 0.0005M $Ca(OH)_2$ solution.

x. What are specific and molar heat capacities? Give their units

xi. Define the four types of crystalline solids on the basis of nature of bonding?

xii. How pressure change affects chemical equilibrium?

xiii. Why heat and work are not state functions?

SECTION –C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) State and explain Hess's Law of constant heat summation.

b) Calculate the oxidation number of Mn in $KMnO_4$, MnO_4,$K_2MnO_3$and Mn_2O_7.

Q4.

a) What is solubility product constant? What is its importance?

b) Calculate the mass of one molecule of oxygen in grams. (Mol. Mass of oxygen = 32g)

Q5.

a) Discuss deviation of gases from ideal behaviour.

b) Explain isomorphism with suitable examples.

Q6.

a) What are Quantum numbers? Discuss Principle and Azimuthal Quantum members in detail.

b) Explain rate law and specific rate constant.